Next When is a matrix RIP?

Deterministic
- Optimal const unknown!

Random
- X has iid sub-Gaussian / Gaussian entries.

$X \in \mathbb{R}^{n \times p}$.

RIP with prob. $1-\varepsilon$.

"Rough Idea" If $n = \Omega\left(s \log \frac{p}{s}\right)$, then a random matrix is RIP wp 0.9999...

$\log \frac{p}{s}$ is the "price" we pay for not knowing where the support is.

$\left(\frac{ep}{s}\right)^s \geq \binom{p}{s} \geq \left(\frac{p}{s}\right)^s$

$\log \binom{p}{s} \sim s \log \frac{p}{s}$

Thm For every $\delta \in (0,1)$, there exists $c_1 := c_1(\delta) > 0$, $c_2 := c_2(\delta) > 0$ such that for every $1 \leq s \leq p$ and $\varepsilon > 0$, if

$$n \geq \frac{c_1 s \log \frac{ep}{s} + \log \frac{1}{\varepsilon}}{c_2}$$

$X \in \mathbb{R}^{n \times p}$

then the matrix $\frac{1}{\sqrt{n}} X$ is (δ, s)-RIP with probability at least $1-\varepsilon$.

with iid $N(0,1)$ entries.

$\left(n = \Omega\left(s \log \frac{p}{s}\right)\right)$

→ Pf will use concentration ⊕ covering numbers.

Remarks:

- $\mathbb{E}\left\|\frac{1}{\sqrt{n}} X\beta\right\|_2^2 = \frac{1}{n} \beta^T \mathbb{E} X^T X \beta = \|\beta\|_2^2$.

  $\frac{1}{\sqrt{n}}$ is the correct scaling.

  (RIP: want to have $\|X\beta\|_2^2 \approx \|\beta\|_2^2$)

  $\left(Y = X\beta^* \iff \frac{1}{\sqrt{n}} Y = \left(\frac{1}{\sqrt{n}} X\right) \beta^*\right)$

- Alternatively, X with $N\left(0, \frac{1}{n}\right)$ entries is (δ, s)-RIP.

$\therefore$ $n = \Omega(s \log \frac{p}{s})$ samples suffice for recovery in the Sparse regression setting.